

Caltech CS156: Learning From Data

Lecture 12: Regularization — Comprehensive Technical Guide

Prof. Yaser Abu-Mostafa — Summarized for Study & Revision

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Abstract

Regularization is the primary curative mechanism for overfitting in machine learning. While VC theory bounds generalization error independent of target properties, overfitting explicitly involves target complexity and noise. This document provides a complete technical analysis of Lecture 12, covering the intuitive, mathematical, and algorithmic foundations of regularization—including the Bias-Variance tradeoff, Weight Decay derivation, Augmented Error, Soft Weight Elimination, Early Stopping, and practical rules for selecting regularizers.

1 Introduction: The Cure for Overfitting

Overfitting occurs when a learning algorithm fits the noise within a finite training sample rather than the underlying target function, resulting in an in-sample error (E_{in}) that decreases while the out-of-sample error (E_{out}) deteriorates.

Sources of Overfitting & The Role of Regularization

Overfitting is driven by two distinct sources of noise:

1. **Stochastic Noise:** Random errors, measurement uncertainties, or non-deterministic fluctuations in target labels y .
2. **Deterministic Noise:** The part of the target function $f(\mathbf{x})$ that cannot be captured by the hypothesis set $\mathcal{H}$ due to model complexity limitations ($f \notin \mathcal{H}$).

Regularization acts as a "handicap" or "brake" on the optimization algorithm, penalizing complex hypotheses to prevent them from fitting both stochastic and deterministic noise.

2 Intuition & The Bias-Variance Tradeoff

To understand how constraining a model improves out-of-sample performance, consider approximating $f(x) = \sin(\pi x)$ with $N = 2$ training points.

Table 1: Empirical Comparison of Unregularized vs. Regularized Linear Model ($N = 2$)

Model Variant	Bias ²	Variance	Total E_{out}	Performance
Unconstrained Line ($h(x) = ax + b$)	0.21	1.69	1.90	Severe Overfitting
Trivial Constant Line ($h(x) = 0$)	0.50	0.00	0.50	High Bias, Zero Variance
Regularized Line (Constrained Slope/Offset)	0.23	0.33	0.56	Optimal Balance

Key Insight: Regularization trades a tiny, acceptable increase in Bias ($0.21 \rightarrow 0.23$) for a massive, dramatic drop in Variance ($1.69 \rightarrow 0.33$), reducing total out-of-sample error from 1.90 down to 0.56.

3 Mathematical Formalization: Weight Decay & Augmented Error

3.1 Legendre Polynomial Transformation

Rather than standard monomials (x^q), we transform input x into a feature space $\mathbf{z}$ using **Legendre Polynomials** $L_q(x)$:

$$\mathbf{z} = \Phi(x) = [1, L_1(x), L_2(x), \dots, L_Q(x)]^T \quad (1)$$

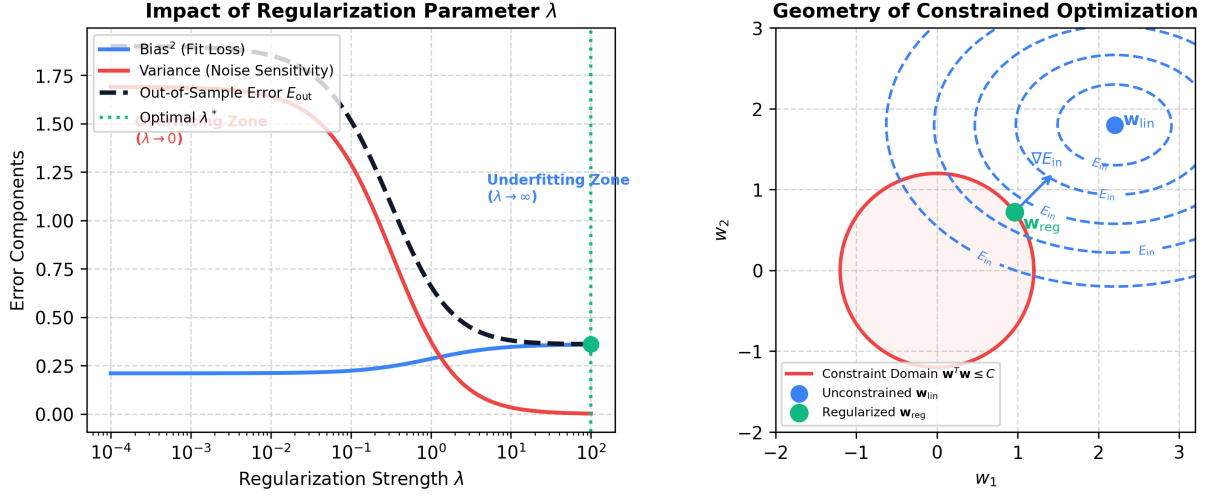


Figure 1: (Left) Bias-Variance tradeoff as a function of regularization strength λ . (Right) Geometric tangency between E_{in} contours and the weight budget constraint sphere $\mathbf{w}^T \mathbf{w} \leq C$.

Legendre polynomials are mutually orthogonal over $[-1, 1]$, making the feature coordinates independent and ensuring that weight magnitude directly correlates with function complexity.

3.2 Constrained Optimization to Unconstrained Augmented Error

The original regularization problem imposes a hard spherical budget C on the weight vector magnitude:

$$\min_{\mathbf{w}} E_{in}(\mathbf{w}) \quad \text{subject to} \quad \mathbf{w}^T \mathbf{w} \leq C \quad (2)$$

Using Lagrange multipliers (KKT conditions), the constrained problem is equivalent to minimizing the **Augmented Error** (E_{aug}) unconditionally:

$$E_{aug}(\mathbf{w}) = E_{in}(\mathbf{w}) + \frac{\lambda}{N} \mathbf{w}^T \mathbf{w} \quad (3)$$

where $\lambda \geq 0$ is the regularization parameter.

- When $C \rightarrow \infty$, $\lambda = 0$ (Unconstrained OLS solution $\mathbf{w}_{lin}$).
- When $C \rightarrow 0$, $\lambda \rightarrow \infty$ (Forced weight decay to $\mathbf{w} = \mathbf{0}$).

3.3 Closed-Form Analytic Solution (Ridge Regression)

Substituting $E_{in}(\mathbf{w}) = \frac{1}{N} \|\mathbf{Z}\mathbf{w} - \mathbf{y}\|^2$ into E_{aug} and differentiating with respect to $\mathbf{w}$:

$$\nabla E_{aug}(\mathbf{w}) = \frac{2}{N} (\mathbf{Z}^T \mathbf{Z} \mathbf{w} - \mathbf{Z}^T \mathbf{y}) + \frac{2\lambda}{N} \mathbf{w} = \mathbf{0} \quad (4)$$

Solving for $\mathbf{w}_{reg}$ yields the regularized pseudo-inverse solution:

$$\mathbf{w}_{reg} = (\mathbf{Z}^T \mathbf{Z} + \lambda \mathbf{I})^{-1} \mathbf{Z}^T \mathbf{y} \quad (5)$$

The addition of $\lambda \mathbf{I}$ guarantees that $(\mathbf{Z}^T \mathbf{Z} + \lambda \mathbf{I})$ is strictly invertible, even when $\mathbf{Z}^T \mathbf{Z}$ is singular or ill-conditioned.

4 Weight Decay in Gradient Descent & Neural Networks

In iterative optimization algorithms such as Gradient Descent (or Backpropagation in Neural Networks), updating weights with augmented error leads to:

$$\mathbf{w}(t+1) = \mathbf{w}(t) - \eta \nabla E_{aug}(\mathbf{w}(t)) = \left(1 - \frac{2\eta\lambda}{N}\right) \mathbf{w}(t) - \eta \nabla E_{in}(\mathbf{w}(t)) \quad (6)$$

Mechanistic Interpretation: At every single step t , the weight vector is first **shrunk** (decayed) by a factor of $\left(1 - \frac{2\eta\lambda}{N}\right) < 1$ before taking the gradient step along E_{in} . This continually pulls weights toward the origin, suppressing high-frequency noise fitting.

5 Taxonomy of Regularization Techniques

Table 2: Overview of Major Regularization Methods

Regularizer Type	Mathematical Penalty $\Omega(h)$	Operational Behavior
Weight Decay (L_2)	$\mathbf{w}^T \mathbf{w} = \sum w_i^2$	Uniformly shrinks all weights toward 0.
Tikhonov Regularization	$\mathbf{w}^T \mathbf{\Gamma}^T \mathbf{\Gamma} \mathbf{w}$	Penalizes specific derivatives/frequencies.
Soft Weight Elimination	$\sum \frac{w_i^2 / \beta^2}{1 + w_i^2 / \beta^2}$	Forces small weights to 0 while leaving large weights alone.
Early Stopping	Implicit via t_{stop} on E_{val}	Limits optimization steps to prevent fitting noise late in training.

6 Practical Rules & Selection Guidelines

1. **Prefer Smooth / Simple Hypotheses:** Both stochastic and deterministic noise are high-frequency and non-smooth. A regularizer that penalizes non-smoothness harms noise fitting significantly more than signal fitting.
2. **Heuristic Choice of $\Omega(h)$, Principled Choice of λ :** Choosing the form of $\Omega(h)$ is often a domain-specific heuristic. However, selecting λ must be strictly principled using **Validation** (Lecture 13).
3. **Safety Net:** If an inappropriate regularizer is chosen, validation will set $\lambda^* \approx 0$, preventing harm to the model.